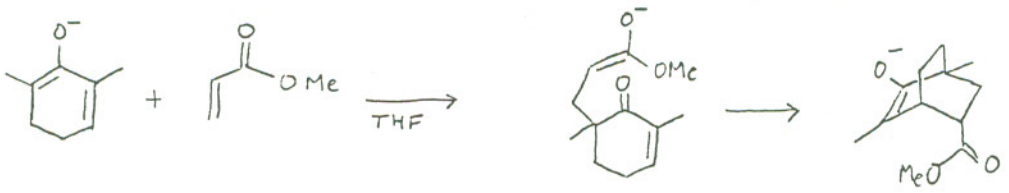
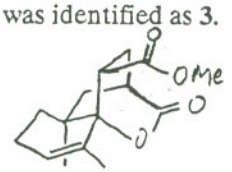
Organic Chemistry Cumulative Exam
November 16, 1991

1. Compound **1**, an important intermediate in the synthesis of a marine sponge sesquiterpene, can be prepared by a double Michael addition between the dienolate **2** and methyl acrylate (Jaafar et al., Tet. Lett. 1991, 32, 5531).



2 **1**

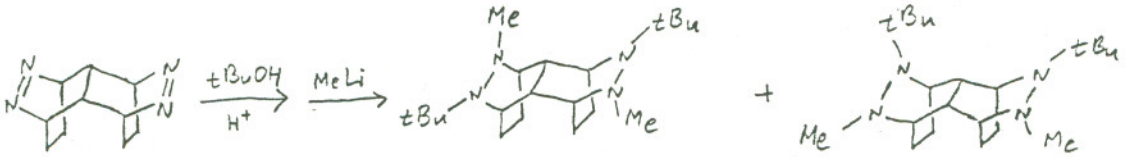
However, a side product was formed, which was identified as **3**.



3

Explain the formation of **3**. How would you minimize this side reaction?

2. Treatment of the bis(diazene) **1** with t-BuOH followed by methyllithium afforded two isomers of the bis(hydrazine), **2a** and **2b** (Nelsen et al., JACS 1991, 113, 7882).



1 **2a** **2b**

The way I've drawn them, the nitrogens in **2a** and **2b** look planar. Of course, this is not the case. The nitrogens are pyramidal and their inversions are slow on the NMR time scale. The ^{13}C NMR spectrum of **2a** shows eight different carbons. The ^{1}H NMR spectrum shows one t-butyl group (δ 1.21) and one N-methyl group (δ 2.62). A NOESY experiment showed that **2a** exhibits strong N-CH_3, central CH cross peaks and extremely weak N-CH_3, CH_2 cross peaks. The ^{13}C NMR spectrum of **2b** shows sixteen different carbons. The ^{1}H NMR spectrum shows two t-butyls (δ 1.18 and 1.24) and two N-methyls (δ 2.55 and 2.63). From this information, determine the conformations of **2a** and **2b**.

3. Bordwell and Ji (JACS 1991, 113, 8398) have measured the acidities of a number of amidines, carboxamides, and thiocarboxamides in DMSO. Some of their data is shown below.

	compound	pK_a
1	$PhCONH_2$	23.35
2	PhCONHPh	18.77
3	$PhC(=NH)NH_2$	26.7
4	PhC(=NPh)NHPh	20.8

Explain how phenyl substitutions increase the acidities of **2** compared to **1** and **4** compared to **3**. Why don't the two phenyl groups in **4** cause a bigger increase in acidity?